

# SHRINIVAS A MASALAVAD

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## PROFILE SUMMARY

AI, Embedded Systems, and Computer Vision engineer with hands-on experience building **8+ AI and embedded engineering projects** across precision agriculture, IoT, wireless networking, and intelligent surveillance systems.

Research Intern at **IIT Dharwad** and Trainee at **NITK Surathkal**, with practical experience in embedded systems, Raspberry Pi, wireless communication, and edge computing.

Presented technology-driven solutions at **8+ national, state, and district-level innovation exhibitions**, including a New Delhi (2024) showcase to researchers, academicians, and government representatives; work recognized by the **Department of Science and Technology (DST), Government of India**.

Skilled in Python, C/C++, OpenCV, YOLO, Raspberry Pi, ESP32, LoRa, IoT, and Edge AI, delivering end-to-end solutions for smart agriculture, wildlife monitoring, and automation — with a strong focus on solving scalable, real-world engineering challenges.

## EXPERIENCE

**Research Intern | Indian Institute of Technology Dharwad (IIT Dharwad)** | Apr 2026 – Present | Dharwad, Karnataka

- Designed and deployed a secure image transmission system between **2 Raspberry Pi devices** operating across different networks.
- Developed a Python-based image transfer pipeline enabling reliable, secure, and low-latency communication between distributed edge devices.
- Configured remote connectivity protocols and optimized network communication for cross-network image transmission.
- Performed **50+ system validation and debugging sessions**, collaborating with faculty researchers to improve communication reliability.

**AI & Embedded Systems Trainee | National Institute of Technology Karnataka (NITK), Surathkal** | May 2023 | Mangalore, Karnataka

- Selected for hands-on research training on an AI-powered Intelligent Weed Recognition and Laser-Based Eradication System for precision agriculture, working under faculty/professor guidance.
- Studied and contributed to the system's architecture and technical design, gaining exposure to simulation, mechanical design, hardware integration, and software development workflows.
- Built foundational experience with ROS 2, Gazebo, and SolidWorks for robotic simulation, system architecture, and virtual testing.

**Technologies:** Python • OpenCV • ESP32 • ROS 2 • Gazebo • SolidWorks • Computer Vision • Embedded Systems

## PROJECTS

### Intelligent Weed Recognition and Laser-Based Eradication System

- Built an AI-powered weed recognition and laser-based eradication system for precision agriculture, trained and evaluated on **1000+ annotated agricultural images**.
- Designed an ESP32-based embedded control system integrating camera and actuator hardware, validated through **50+ prototype experiments** using an end-to-end Python/OpenCV computer vision pipeline.
- Delivered a functional precision-agriculture solution combining AI-based weed detection with automated laser eradication for real-world field use.

### Smart Laser-Based Elephant Intrusion Detection System — KSSC / DST, Government of Karnataka

- Designed a laser-based virtual perimeter system for early elephant intrusion detection in coffee plantations and farmland.
- Built an embedded monitoring system with ESP32, Laser/IR sensors, and **3 communication modules (LoRa, GSM, Wi-Fi)** for real-time alerts across **2 channels** (SMS and mobile app).
- Selected through KSSC evaluation to present at the Karnataka State-Level Science Exhibition, engaging international delegates.

### Smart Solar-Powered Bird Repellent System for Crop Protection — Embedded Systems / Renewable Energy

- Built a solar-powered autonomous bird repellent system with a **6000 mAh battery** and time-based automation (5:00 AM–6:00 PM), cutting unnecessary power consumption by **~11 hours/day**.
- Programmed rotating deterrent-sound sequences every 30 minutes to reduce bird habituation and improve effectiveness.

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++, SQL

**AI & Computer Vision:** Machine Learning, Deep Learning, Computer Vision, OpenCV, YOLO, Image Processing, Edge AI

**Embedded Systems & IoT:** ESP32, Raspberry Pi, Arduino, Embedded C, IoT, Sensor Integration, GPIO, UART, I2C, SPI

**Robotics & Simulation:** ROS 2, Gazebo, SolidWorks, Robotics Fundamentals

**Networking & Edge Computing:** TCP/IP, SSH, Remote Connectivity, Wireless Communication, LoRa, Edge Computing

**Tools, Frameworks & Cloud:** Git, GitHub, Linux, VS Code, Jupyter Notebook, Firebase, MySQL (Basic)

## EDUCATION

**Master of Computer Applications (MCA)** | Expected 2028

Bengaluru University, Bengaluru | Pursuing

**Bachelor of Computer Applications (BCA)** | Graduated 2026

Karnataka University, Dharwad | CGPA: 8.3/10

## CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Specialization** — Coursera (DeepLearning.AI & Stanford University): Supervised Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommender Systems, Reinforcement Learning.
- Selected for **2 National-Level, 2 State-Level, and 4 District-Level Science & Innovation Exhibitions (8+ total)**.
- Presented projects alongside international delegates (Germany, Nigeria), fostering technical knowledge exchange.
- Innovation project appreciated by the **Department of Science and Technology (DST), Government of India** for real-world impact.